

Valorization of coffee cherry waste ash as a sustainable construction material

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Highlights

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Coffee cherry waste (CCW), a residue of coffee processing, was treated and used as a partial replacement for cement.

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Incorporating treated CCW up to 15 % replacement of cement improved the [compressive strength](#) of mortar at all curing ages.

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Hydration studies showed that treated CCW enhanced [cement hydration](#) and led to a denser microstructure.

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Sustainability assessment revealed that incorporating treated CCW significantly improved the environmental performance.

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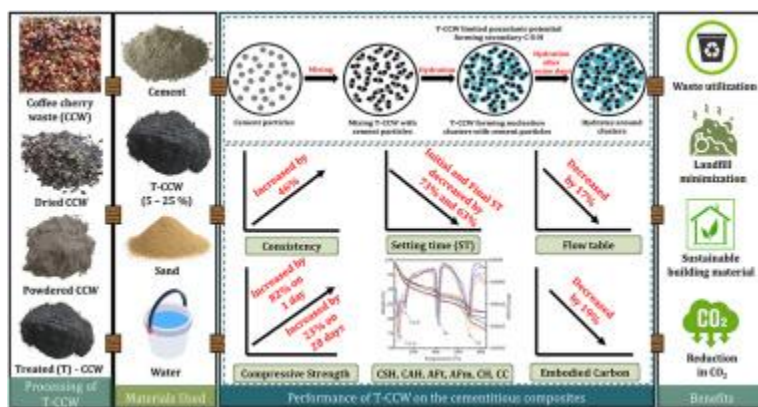
A fishbone diagram was developed to identify the potential challenges in the development and application of treated CCW.

Abstract

This study explores the potential of treated coffee cherry waste (T-CCW) as a partial replacement of cement in mortar. T-CCW was characterized and incorporated into pastes and mortars at 5 %–25 % cement replacement. The main objectives were to examine the fresh and hardened properties, hydration, and environmental assessment. Results showed

that the high specific surface area and [porous structure](#) of T-CCW particles increased water demand and accelerated setting times. T-CCW incorporation of up to 15 % enhanced [compressive strength](#) at all curing ages due to improved hydration and limited [pozzolanic reactions](#). Ultrasonic [pulse velocity](#) indicated good homogeneity and compactness in T-CCW blended mortars. [Microstructural analysis](#) revealed that T-CCW enhanced [cement hydration](#), leading to a denser matrix. Environmental analysis showed a reduced [embodied carbon](#) and cement intensity index compared to the control mix. Overall, the [optimal performance](#) was observed at 15 % T-CCW replacement, significantly improving engineering properties and environmental impact. Further, the fishbone diagram addresses various factors to optimize the use of T-CCW as a [cementitious composite](#). These findings demonstrate the potential of T-CCW as a [sustainable construction material](#), offering a promising pathway towards environmentally friendly and resource-efficient building practices while addressing waste management in the coffee industry.

Graphical abstract



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Introduction

The excessive production of anthropogenic carbon dioxide (CO₂) emissions seriously threatens ecosystems, and the damage caused by these emissions is immutable [1]. Anthropogenic activities that lead to this problem involve various processes, such as the use of fossil fuels for energy generation and the production of ordinary Portland cement (OPC) [2]. The production of OPC is responsible for approximately 7–8% of worldwide greenhouse gas emissions, making it one of the construction industry's most environmentally impactful materials [3]. Various strategies have been employed to reduce CO₂ emissions from cement production: a) improving energy efficiency, b) substituting fuels, c) implementing carbon capture and storage technologies, d) promoting effective

cement usage, and e) using supplementary cementitious materials (SCMs) [4,5]. The improvement in cement production through certain strategies may be limited due to the need for specialized equipment requirements, leading to higher costs [6,7]. However, using SCMs as a replacement for OPC provides a viable alternative due to the economic benefits of using locally sourced materials and the potential for large-scale production. Industrial and agricultural wastes are feasible alternatives for cement binders.

Agricultural waste ashes have been extensively studied as SCMs to enhance the sustainability and performance of cement-based materials. Various sources, including walnut shell ash (WSA) [8], oat husk ash (OHA) [9], pistachio shell ash (PSA) [10], banana leaf ash (BLA) [11], wheat straw ash (WSA) [12], barley straw ash (BSA) [13], palm oil fuel ash (POFA) [14], rice straw ash (RSA) [15], bagasse ash (BA) [16], rice husk ash (RHA) [17], corn cob ash (CCA) [18], corn stover ash (CSA) [19], etc., have shown promising results. These materials exhibit diverse effects on cement properties. For instance, WSA improves pozzolanic activity, setting time, and long-term strength by promoting fibrous calcium silicate hydrate (C-S-H). OHA, calcined at 600 °C, can replace up to 20 % of OPC without compromising strength. PSA enhances workability, setting time, and at 10 % replacement, yielding a 17 % increase in strength. BLA and WSA contribute to increased strength, with BLA reducing water absorption and WSA enhancing the microstructure of cement concrete. BSA optimizes the pore structure of concrete, leading to significant improvement in strength up to 15 %. POFA in nano size can improve the compressive strength up to 30 % without any reduction in strength. RSA, BA, and RHA contain significant amounts of silica, improving workability, strength, and durability. Untreated CCA presents challenges due to its high potassium content, but pretreated CSA enhances silica content, accelerating hydration and improving strength. The environmental benefits of these SCMs are significant, reducing landfill waste, lowering CO₂ emissions, and recycling agricultural waste into valuable construction material. However, each material's performance depends on its composition, processing conditions, and percentage of cement replacement, highlighting the need to optimize their application in cementitious composites. While significant progress has been made in this research, many agricultural wastes remain underutilized. This research explores coffee cherry waste (CCW) as a novel, sustainable alternative for construction materials, contributing to the enduring efforts to develop more environmentally friendly building practices.

The utilization of agricultural waste proves to be a cost-effective alternative compared to industrial waste [20]. Despite its application in various industries, disposing of agricultural waste through burning in open fields or using methods like stacking or landfilling can negatively impact the environment. Improper waste management and by-products can lead to soil degradation and water pollution. Also, landfill disposal leads to methane gas

emissions, which are significantly more potent than CO₂, regarding their impact on global warming [21]. Thus, recycling and reusing waste can benefit the environment, economy, and society [22].

Globally, coffee is considered a highly significant commodity, widely traded, and the most consumed beverage [23]. The coffee industry significantly impacts the global economy, with Brazil, Vietnam, Colombia, Indonesia, India, Ethiopia, and Honduras being the biggest coffee producer countries, representing about 80 % of global production [24]. It is concerning to note that coffee consumption and waste generation have increased simultaneously. During coffee production, a considerable amount of waste is generated, with 45 % of the coffee plantation left as residue [25]. Also, during the washing process, a large quantity of water becomes contaminated and carries excessive carbon, which can harm the environment [26]. The technique used to process coffee can generate by-products throughout the various stages of processing, including pre-roasting coffee, dry processing (coffee cherry waste), semi-dry and wet processing (coffee pulp), and post-roasting coffee (coffee silverskin and spent coffee grounds) [27]. Fig. 1 shows the steps involved in the coffee processing using the wet post-harvesting method.

Coffee cherry waste (CCW) is the non-useable part of the coffee beans and is the by-product primarily obtained through the wet method during coffee processing [28]. These wastes cover the coffee beans for approximately 12 % of the cherry's dry weight [29]. It has been reported that for every tonne of coffee processed, the coffee processing industry produces approximately 0.18 tonnes of CCW [30]. The coffee cherries are taken to a processing facility where the under-ripe and overripe cherries are separated and treated accordingly. The seeds are then extracted from the cherries using a de-pulping machine. Approximately 90 % of the coffee cherry waste is often discarded into a compost pile or dumped in landfills or water bodies. The improper disposal of these coffee cherry waste poses significant environmental problems for the soil and water around the farmland due to its high acidity and high levels of caffeine, tannins, and other polyphenols [31]. These wastes can acidify the soil, making nutrients unavailable for crops. Additionally, the long-term release of caffeine into water bodies negatively impacts the aquatic environment. Further, decomposing coffee cherries releases toxic mycotoxins that grow on coffee cherries, causing negative effects on the central nervous, cardiovascular, and respiratory systems [32]. The steps involved in generating CCW are shown in Fig. 2.

With an estimated increased production of coffee in the future, waste management policies for CCW must be implemented for proper handling and application, contributing to a circular and sustainable economy. Studies have investigated the possibility of using spent coffee grounds, but significant research has yet to be done on utilizing CCW as a

construction material. This motivated the exploration of the potential of CCW as an alternative construction material to address the significant environmental challenges associated with cement production and the improper disposal of this byproduct.

According to the literature, more research is needed on the possible utilization of CCW as a substitute for cement. It is clear from previous studies that processing methods like burning and grinding significantly impact reactivity [33]. Hence, this research explores the possibility of using treated coffee cherry waste (T-CCW) for normal-strength mortar application. The following objectives were studied to investigate the feasibility of utilizing T-CCW as a construction material.

- i)

To perform an in-depth material characterization of T-CCW.

- ii)

To investigate the effect of T-CCW on cement paste and mortar properties.

- iii)

To study the effect of T-CCW on the hydration of cement paste.

An extensive experimental program has been carried out to answer these objectives. Different microstructural analyses are performed to investigate the characterization of T-CCW by a) X-ray diffraction (XRD), b) Thermogravimetric analysis (TGA), and c) Scanning electron microscopy (SEM), d) Energy dispersive spectroscopy (EDS) e) Fourier transform infrared spectroscopy (FTIR). Modified Chapelle tests were performed to assess the pozzolanicity. For the fresh properties of the T-CCW paste, tests such as a) consistency, b) setting time, and c) mini-slump flow are considered. The study investigates the hardened properties of T-CCW mortar. The hydration characteristics of T-CCW paste are conducted using SEM, XRD, TGA, and FTIR. A fishbone diagram is drawn to identify, analyze, and display the potential causes of using T-CCW in cement.

Cement and fine aggregate

The commercially available OPC of 53 grade conforming to IS 12269 is utilized in the present study. The chemical composition of OPC is shown in Table 1. The normal consistency is 29 %, the specific gravity is 3.14, 160 min initial setting time and 310 min final setting time, and the specific surface area of OPC was 365 m²/kg. Natural river sand was used as a fine aggregate. The sand was washed, sundried, and sieved. The fine aggregate used in this study is categorized as zone 2, passing through

Characterization of T-CCW

The morphology of the T-CCW is studied by using SEM, as shown in Fig. 6(a). T-CCW particles exhibit irregular shapes with wrinkled and rugged surfaces [31,41]. Further, T-CCW particles exhibit the presence of numerous pores on the surface, indicating the release of volatile and organic matter from the untreated CCW during the thermal decomposition process. These pores can absorb water and serve as an internal curing agent. The SEM of the untreated CCW is illustrated in Fig. S1. The EDS analysis

Scanning electron microscope

Fig. 14 presents the SEM images of C0, C10, and C25 pastes after 7 days. The SEM images show the formation of hydration products, such as the needle-shaped ettringite (Aft) crystals, distinctive hexagonal-prismatic crystals of portlandite (CH), and C-S-H gel. The T-CCW samples (C10 and C25) showed traces of monosulphate (AFm). The T-CCW admixed pastes exhibit a dense microstructure compared to control pastes, resulting in increased hydration products. The hydration process involves pore

Sustainability assessment

Researchers are exploring various alternative materials derived from industrial by-products and agricultural wastes in the search for sustainability in the construction industry. These materials not only substitute for conventional materials but also significantly lower the overall carbon footprint of the production. This approach addresses the pressing need for sustainable construction practices and tackles waste management challenges.

A simplified methodology is adopted to quantify the

Factors affecting the utilization of T-CCW as a sustainable construction material

The fishbone diagram is a tool used to identify and explore the potential causes of a specific problem. It also provides ideas for future work to enhance further understanding of the problem. Fig. 17 presents a fishbone diagram illustrating the factors influencing the performance of CCWA as a cementitious composite. This visual representation summarizes the various parameters contributing to the effectiveness of CCWA in construction applications. The diagram outlines eight primary factors: raw

Conclusion

Reducing, reusing, and recycling are crucial to achieving sustainability. Exploring alternative materials from agricultural waste, such as CCW, generated from coffee processing addresses waste management challenges while decreasing the use of energy-intensive materials. This study examines the potential of T-CCW as a supplementary cementitious material, contributing to the development of sustainable construction practices. The following conclusions can be drawn from the study.

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The addition of

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CRediT authorship contribution statement

Balasubramanya Manjunath: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Claudiane M. Ouellet-Plamondon:** Writing – review & editing, Validation, Data curation, Conceptualization. **Anjali Ganesh:** Writing – review & editing, Funding acquisition. **B.B. Das:** Writing – review & editing, Validation, Data curation. **Chandrasekhar Bhojaraju:** Writing – review & editing, Writing – original draft,

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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